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Association between loneliness and suicidality among general populations and persons with depressive and bipolar disorders: A systematic review

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ABSTRACT

Background: Loneliness and suicidality are interrelated phenomena. Several studies suggest that they often co-exist, but the magnitude is unclear. This systematic review aims to analyze this association in general population samples of varying age groups and persons with depressive and bipolar disorders.

Methods: Search terms in this review included words related to loneliness and suicidality among general populations and depressive and bipolar disorders. Databases included PubMed, PsychINFO, and Web of Science. The search occurred up until May 27, 2024. Screening and data extraction were performed independently. Studies were categorized by age group or clinical diagnosis. Quality assessments were conducted using NIH tools.

Results: Fifty-six studies met eligibility criteria: 52 involved general population samples and 4 involved patients with depressive or bipolar disorders. In healthy adults aged 18 to 64, loneliness mainly showed a moderate positive correlation with suicidality ($r = 0.26$ to 0.59), while correlations in older adults (aged 65+) ($r = 0.498$) and in adolescents aged 13-17 were weaker. In depressive and bipolar disorders, correlations ranged from weak to moderate ($r = -0.06$ to 0.40), with associations stronger in unadjusted models (UOR = 2.8 to 7.07). Furthermore, some studies suggested that depressive symptoms mediate the association between loneliness and suicidality.

Conclusion: A moderate and positive association was observed between loneliness and suicidality in healthy adults and patients with depressive disorders. However, the role of bipolar disorders in the association remains unclear. Practitioners should routinely evaluate persons living with depressive or bipolar disorders for loneliness as part of a comprehensive assessment.

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1. Introduction

Suicidality encompasses a range of behavioural dimensions, including suicidal ideation and attempts, which serve as non-fatal risk factors for completed suicide (Baiden et al., 2018; Dev et al., 2024; Kurniawan et al., 2024; Putra et al., 2019). The forgoing suicidal behaviours contribute to a global public health crisis, where completed suicide is one of the leading causes of death. Consequently, improving suicide prevention services and resources that account for various risk factors for suicidality is a priority.

Understanding the effect of risk factors on suicidal behaviours is critical for suicide prevention (Bommersbach et al., 2022; Lu et al., 2020; Park et al., 2020). Extant literature has shown that psychosocial factors, such as inadequate perceived social support, inadequate social behaviour, and loneliness are risk factors for suicidality (Bommersbach et al., 2023a; Bommersbach et al., 2023b; Na et al., 2022; Park et al., 2020; Yang et al., 2023a). In particular, loneliness is a frequent issue experienced by all age groups and clinical populations. Loneliness is described as an emotion derived from personal experiences of social isolation, resulting from inadequate or unfulfilling social connections (Gomboc et al., 2022). It is crucial to recognize that loneliness depends on the perception and therefore evaluation of one's social connections according to one's standards, not the number of relationships one has. According to the World Health Organization (WHO) (2025), globally, approximately 1 in 4 older adults experience loneliness and between 5 and 15 % of adolescents experience loneliness. Behaviours associated with loneliness include but are not limited to social withdrawal and vigilance for social threats (Hawkley and Cacioppo, 2010). Although loneliness is a multifaceted emotion, feelings of loneliness and perceptions can change (Hansen et al., 2024). Targeting the causes of loneliness through improving social skills, changing maladaptive thinking patterns, and promoting social engagement can help reduce feelings of loneliness. Thus, incorporating loneliness reduction into suicide prevention services and strategies has the potential to improve overall mental health outcomes and reduce the likelihood of suicidal ideation and behaviours (Jobes et al., 2015; Aboagye et al., 2022).

Although loneliness is a risk factor for suicidal behaviours, it is essential to consider age in the association as the experience, causes, and effects of loneliness can differ considerably across various age groups, including adolescent, adult, and older adult populations (Lee, 2023; Wilkialis et al., 2021). The differences in loneliness across age groups can be attributed to factors such as social and emotional development, and age-specific life circumstances and changes (Franssen et al., 2020; Manoli et al., 2022). Therefore, effective suicide prevention should consider age as a moderator when reducing loneliness as a risk factor (Motillon-Toudic et al., 2022). In addition to age, depression moderates the relationship between loneliness and suicidality, potentially altering the strength or direction of their association (Oh et al., 2024b; Ribeiro et al., 2018). Loneliness has been associated with an increase in depression, which can, in turn, elevate the risk of suicidality. Understanding the extent to which age and depression influences the association between loneliness and suicidality may elucidate why loneliness may more strongly contribute to suicidality in certain individuals (Motillon-Toudic et al., 2022; Yang et al., 2023b).

Herein, although previous studies have examined the relationship between loneliness and suicidality, this systematic review focuses on age as a potential moderator in general population samples. Moreover, we explore the interaction this association has with depressive symptoms and depressive and bipolar disorder. We also compare findings from the general population and depressive and bipolar studies to provide further insights.

2. Methods

2.1. Search strategy

This review follows the 2020 Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) guidelines (Page et al., 2021). A comprehensive literature search was conducted using databases including PubMed, PsychINFO, and Web of Science. The search string was inserted into each database. Further details can be found in the supplementary materials (Table S1). All terms relevant to loneliness, suicidality, and characteristics of general population samples and patients with depressive and bipolar disorders were used. The literature search occurred from inception to May 27, 2024.

2.2. Study selection and eligibility criteria

Covidence was employed to facilitate the study selection process (Covidence, 2024). Two independent reviewers (A.L. and A.C.) conducted an initial screening of abstracts and titles for potentially relevant studies. Subsequently, full-text screening was performed by the same reviewers. Only articles deemed relevant according to the eligibility criteria, by both reviewers, were included for data extraction and analysis (Table 1). Any conflicts were resolved between the two reviewers or by consulting a third reviewer. Covidence automatically eliminated many duplicate studies. Additional duplicates were manually removed.

Studies were eligible for inclusion if they were primary articles [except randomized control trials (RCTs)] investigating the association between loneliness and suicidality (ie., suicidal behaviour, ideation, planning, attempt, etc.) in general population samples aged 13 or above or persons diagnosed with depressive and bipolar disorders [i.e., major depressive disorder (MDD), bipolar disorder (BD), or persistent depressive disorder (PDD)]. Additionally, only retrievable full-text articles in English were considered. Studies were excluded if they were secondary sources, were conducted about or during the COVID-19 pandemic without specifying the participants' health status, or involved clinical populations with comorbidities or mixed diagnoses. Case studies and animal studies were also excluded (Table 1).

Table 1
Eligibility criteria.

Inclusion Criteria	1. Primary research (i.e., cross-sectional studies, longitudinal studies, case-control studies, secondary analyses), 2. Studies that describe an association between loneliness and suicidality, 3. Participants that are healthy older adults (≥65), adults (18–64) and adolescents (13–17), or patients with or have a high likelihood of depressive or bipolar disorders (major depressive disorder, bipolar disorder, persistent depressive disorder, etc.), 4. Patients with depressive or bipolar disorders must be diagnosed through the DSM or ICD, 5. Full-text articles available and in English.
Exclusion Criteria	1. Non-primary research (i.e., literature reviews, systematic reviews, meta-analyses, posters, abstracts, guidelines, protocols, theses, books, opinion pieces, and non-academic reports), 2. Randomized control trials, 3. No description of an association between loneliness and suicidality, 4. Participants from a clinical population with conditions other than depressive or bipolar disorders, including comorbidities and mixed diagnoses, 5. Participants with an unidentified health status in COVID studies, 6. Participants who have ended their life, 7. Case studies and animal studies, 8. Full-text unavailable or not in English.

2.3. Data extraction

Three reviewers (A.L., A.C., and T.M.) independently conducted data extraction on included studies. Information to be extracted was established a priori and included: 1) the first author and year of publication, 2) study type, 3) sample population and size, 4) age range, 5) assessment tools used for measuring loneliness and suicidality, and 6) outcomes of interest (Table 2). All results that examined the association [i.e., correlations, unadjusted odd ratios (UOR), adjusted odd ratios (AOR), beta-coefficients, variance, etc.] between loneliness and suicidality was considered an outcome of interest. Studies were organized by their sample population, distinguishing between general population samples and patients diagnosed with depressive or bipolar disorders. Within the general population studies, studies were further categorized according to age groups: adolescents (13–17 years old), adults (18–64), older adults (≥65), adults and adolescents (13–64), adults and older adults (≥18), and adolescents, adults, and older adults (≥13). Studies involving patients with depressive or bipolar disorders were categorized according to the specific diagnosis (MDD, BD, or PDD).

2.4. Quality assessments

Studies were critically appraised by two independent reviewers (A.L. and T.M.) using the established tools provided by the National Institutes of Health (NIH), including the Quality Assessment of Observational Cohort and Cross-Sectional Studies, the Quality Assessment of Case-Control Studies, and the Quality Assessment Tool for Before-After (Pre-Post) Studies with No Control Group Studies (Ma et al., 2020; NIH, 2021). Detailed risk of bias assessment tables are available in the supplementary materials (Tables S2-S4). Comprehensive details on items and rating can be found on the NIH website (NIH, 2021).

2.5. Comparative analysis

Associations (correlations and odd-ratios) in general population samples were compared with studies with patients with depressive or bipolar disorders. General trends in findings were used to make comparisons where appropriate. In addition, age groups separated in general population samples were compared with each other. Studies separated by depressive and bipolar disorders were also compared with each other.

3. Results

3.1. Search results

The search for relevant literature yielded 4912 studies. After the removal of duplicates ($n = 2629$), 2283 studies were left. Screening was based on the eligibility criteria (Table 1). After abstract and title screening, 2046 studies were excluded and 237 studies moved onto full-text screening. Subsequently, full-text screening excluded 181 studies, leaving 56 studies for data extraction and analysis. Reasons for full-text exclusion were due to the full-text not being available ($n = 45$), not in English (n = 4), non-primary research ($n = 8$), wrong outcomes ($n = 24$), wrong study design (n = 2), and wrong patient population ($n = 98$). Additional details on the study selection process can be found in the PRISMA flow diagram (Fig. 1).

3.2. Methodological quality assessment

All cross-sectional and observational cohort studies clearly stated their sample population and examined different levels of loneliness as related to suicidality. Approximately half the studies accounted for confounders. All cross-sectional studies measured loneliness and suicidality at one point in time and could not measure suicidality before the outcome. In addition, most studies did not report a sample justification and most had a participation rate of above 50 %. There were 5 studies with a high risk of bias, 44 with moderate risk, and 4 studies with low risk. The studies with high risk of bias include Garcia-Williams et al. (2014), Keiner et al. (2023), Pereira and Cardoso (2018), Rich and Bonner (1987), and Weber et al. (1997).

There were two case-control studies, Pike et al. (2024) and Wiktorsson et al. (2010), both of which had a low risk of bias. There was no sample size justification for both studies. All eligible subjects participated in the studies and it was impossible to measure exposure (loneliness) before the outcome (suicidality). Assessors could not be blinded to the loneliness status of cases and controls. In addition, Wiktorsson et al. (2010) did not use concurrent controls, while Pike et al. (2024) did not report use of concurrent controls.

Only one study, Wolf et al. (2022), had a pre-post with no control group design. This study had a low risk of bias, despite not reporting if the sample size was large enough to present confidence in the findings. In addition, the study did not collect outcome measures multiple times before and after the intervention.

3.3. General population samples

Cross-sectional and observational cohort studies consistently demonstrated a significant and positive association between loneliness and suicidality among general populations, despite heterogeneity in study designs, including sample population characteristics and scales used to measure both variables (Alpass and Neville, 2005; Bennardi et al., 2017; Bonner and Rich, 1987, 1990; Chang et al., 2010, 2015, 2017a, 2017b, 2017c, 2019; Chan et al., 2016; Chesin and Jeglic, 2012; Cleverley and Kidd, 2011; Dema et al., 2019; Durak Batigün, 2005; Ernst et al., 2021; Garcia-Williams et al., 2014; Garnefski et al., 1992; Gomboc et al., 2022; Joiner Jr and Rudd, 1996; Keiner et al., 2023; Kim et al., 2022; Kirkpatrick-Smith et al., 1992; Klugman et al., 2024; Lagman et al., 2021; Lamis et al., 2014; Lamlé et al., 2023; Larson et al., 2024; Lasgaard et al., 2011a, b; Li et al., 2016; Marthoenis and Nassimbwa, 2022; McClelland et al., 2023; McKinnon et al., 2016; Muyan and Chang, 2015; Oh et al., 2024a; Ofei-Dodoo et al., 2020; Page et al., 2006; Peltzer and Pengpid, 2017; Pereira and Cardoso, 2018; Pitman et al., 2020; Putra et al., 2019; Rich and Bonner, 1987; Sarfo et al., 2023; Shirima et al., 2023; Stravynski and Boyer, 2001; Vuscan et al., 2022; Weber et al., 1997; Yang et al., 2023a; Yadegarfard et al., 2014; Zhang et al., 2021; Zhou et al., 2023). Across studies, populations originated from the United States, Europe (e.g., Turkey, Hungary, Germany, and Denmark), and Asia (e.g., China, Indonesia, Thailand, and the Philippines). UCLA Loneliness Scale was most commonly used to measure loneliness. Its reliability and adaptability to various cultural contexts allowed accurate cross-study comparisons (Alsubheen et al., 2021). In contrast, many shorter, single-item measures were also employed, which lacked the evidence of reliability and validity needed to make accurate comparisons between studies. Several scales were used to assess suicidality, including the Suicidal Behaviours Questionnaire (SBQ) and the Beck Hopelessness Scale (BHS). These scales were well-validated and provided comprehensive assessments (Amini-Tehrani et al., 2020; Balsamo et al., 2020). Some studies also used less extensive scales or single-item assessments from surveys, sharing the same issues as aforementioned limited scales measuring loneliness. Most general population studies were cross-sectional and rated fair in quality assessments.

In the category of healthy older adults (≥65 years old), Alpass and Neville (2005) was the sole eligible study. They reported a significant, positive, and moderate correlation between loneliness and suicidality, $r = 0.498$, suggesting that loneliness may be a substantial risk factor for suicidality among healthy older adults.

Comparatively, studies involving healthy adults (18–64 years old) found significant, positive, and moderate correlations ranging from $r = 0.26$ to 0.59, indicating that while the association was generally consistent across all 19 studies, the strength of the association varied depending on sample characteristics and measures used for loneliness and suicidality (Bonner and Rich, 1987, 1990; Chang et al., 2010, 2015,

Table 2
Sample characteristics for studies examining the association between loneliness and suicidality in general population samples and patients with depressive and bipolar disorders.

Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
Healthy adolescents, adults, and older adults (≥13)					
Durak Batigün (2005)	Cross-sectional	$n = 683$ Individuals residing in Ankara and İzmir in Turkey	15–65	Loneliness -The UCLA Loneliness Scale (UCLA-LS) Suicidality -Suicide Probability Scale (SPS)	Loneliness had a significant positive correlation with suicide probability ($r = 0.60$, $p < 0.001$).
Stravynski and Boyer (2001)	Cross-sectional	$n = 19{,}724$ Data drawn from the Quebec Health Survey conducted in 1987 on Quebec residents living in private households	≥15	Loneliness - Three items measuring loneliness Suicidality - Single item to measure parasuicide - Single item to measure suicide ideation	Odds ratios were calculated for one-year prevalence of suicide ideation and parasuicide by level of loneliness. ● Suicide Ideation ○ Feeling lonely very often ($n = 18{,}384$): UOR(CI 95 %) = 10.5 (8.4–13.1) ● Parasuicide ○ Feeling lonely very often ($n = 17{,}972$): UOR(CI 95 %) = 13.5 (9.3–19.6)
Healthy older adults (≥65)					
Alpass and Neville (2005)	Cross-sectional	$n = 217$ Old residents from a small New Zealand city	65–89	Loneliness -Revised UCLA Loneliness Scale (R-UCLA) Suicidality -The Suicidal Ideation Questionnaire	Suicidal ideation was found to have a significant positive correlation with loneliness ($r = 0.498$, $p < 0.001$).
Healthy adults (18–64)					
Bonner and Rich (1990)	Cross-sectional	$n = 146$ Male inmates from Palm Beach County, Florida, Detention Facility	M(SD): 28.28 (6.77)	Loneliness - UCLA Loneliness Scale Suicidality - Self-report version of the Scale for Suicidal Ideation (SSI)	Correlational analysis showed that there was a positive and significant correlation between suicidal ideation and loneliness ($r = 0.44$, $p < 0.05$). Using a multiple regression analysis, stepwise forward-inclusion algorithm, low reasons for living, irrational beliefs, jail stress, and loneliness combined accounted for 51 % of the variance in suicidal ideation. Loneliness: ($\beta = 0.15$, $p < 0.05$; $R = 0.71$, $F(4, 141) = 36.81$, $p < 0.05$). In a hierarchical multiple-regression analysis, loneliness, jail stress, and loneliness × jail stress were entered. Loneliness was positively and significantly associated with suicide ideation. Jail stress moderated the association. ● Loneliness: $R = 0.44$, β: 0.44, t: 5.75, $p < 0.05$ ● Jail Stress: $R = 0.53$, β: 0.32, t: 4.24, $p < 0.05$ ● Loneliness × Jail Stress: $R = 0.56$, $\beta = 0.16$, t: 2.27, $p < 0.05$
Bonner and Rich (1987)	Cross-sectional	$n = 158$ Participants were from the general psychology subject pool at Indiana University of Pennsylvania	Estimate: 18–22	Loneliness - UCLA Loneliness Scale Suicidality - Suicidal Behaviours Questionnaire (SBQ)	Correlational analysis showed that there was a positive and significant correlation between suicidal ideation and loneliness ($r = 0.26$, $p < 0.05$). In stepwise multiple regression analysis, loneliness, irrational beliefs, and reasons for living explained 15 % of the variance in suicidal behaviour. Loneliness alone was positively and significantly associated with suicidal behaviour ($R = 0.24$, $R^2 = 0.06$, $\beta = 0.24$, $p < 0.05$). The addition of irrational beliefs and reasons for living to loneliness (step 1) added explained more variance.
Chang et al. (2010)	Cross-sectional	$n = 160$	18–46	Loneliness - UCLA Loneliness Scale	Correlation analysis between UCLA and SBQ-R revealed a significant positive

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Table 2 (*continued*)

Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
		Hispanic adults studying at a large public college in the East Coast of the United States		Suicidality - Suicidal Behaviours Questionnaire-Revised (SBQ-R)	correlation ($r = 0.49$, $p < 0.001$). In hierarchical regression analyses, UCLA significantly accounted for 24 % of the variance in SBQ-R. Adding step 2 LECCS-NEG (negative life event measurement) contributed an additional 2 % but non-significant variance in outcome. The interaction term of UCLA-R and LECCS-NEG accounted for an additional significant 3 % of the variance in suicidal behaviours. Overall, the model explained a significant 29 % of the variance in suicidal behaviours in Hispanics and demonstrated that LECCS-NEG moderated the association between loneliness and suicidal behaviours by strengthing it the more negative life events an individual goes through. ● Step 1: UCLA-A, $\beta = 0.49$, $R^2 = 0.24$, $p < 0.001$ ● Step 2: LECCS-NEG, $\beta = 0.14$, $R^2 = 0.26$, p = NS ● Step 3: UCLA-R × LECCS-NEG, $\beta = 1.05$, $R^2 = 0.29$, $p < 0.05$
Chang et al. (2015)	Cross-sectional	$n = 334$ College students from a university in the southeast of United States	18–58	Loneliness - Revised UCLA Loneliness Scale (R-UCLA) Suicidality - Single item from Suicidal Behaviours Questionnaire-Revised (SBQ-R)	In correlational analysis, suicide probability and loneliness had a positive and significant association ($r = 0.46$, $p < 0.001$). In hierarchical regression analysis with suicide probability as the outcome and loneliness and sexual assault as predictors: ● Step 1: Loneliness, $\beta = 0.42$, $R^2 = 0.17$, $p < 0.001$ ● Step 2: Sexual Assault, $\beta = 0.15$, $R^2 = 0.19$, $\Delta R^2 = 0.02$, $p < 0.01$ ● Step 3: Loneliness × Sexual Assault, $\beta = 0.52$, $R^2 = 0.20$, $\Delta R^2 = 0.01$, $p < 0.05$ This suggests that the association between loneliness and suicide probability is stronger for individuals who have experienced sexual assault.
Chang et al. (2017a)	Cross-sectional	$n = 457$ Hungarian college students from a public university in Budapest, Hungary	18–32	Loneliness - Revised UCLA Loneliness Scale (R-UCLA) Suicidality - The Frequency of Suicidal Ideation Inventory (FSII)	There was a significant positive association between R-UCLA and FSII ($r = 0.59$, $p < 0.001$). In a hierarchical regression analysis examining suicidal ideation as the outcome variable, step one included loneliness as a predictor, yielding $\beta = 0.59$. Loneliness explained 34 % of the variance in suicidal ideation and was statistically significant. This association was negative when the interaction term loneliness × optimism was added, suggesting that optimism moderates the association between loneliness and suicidal ideation. ● Step 1: Loneliness: $\beta = 0.59$, $R^2 = 0.34$, $p < 0.001$ ● Step 2: Optimism: $\beta = -0.16$, $R^2 = 0.36$, $\Delta R^2 = 0.02$, $p < 0.001$ ● Step 3: Loneliness × Optimism: $\beta = -0.57$, $R^2 = 0.40$, $\Delta R^2 = 0.04$, $p < 0.001$
Chang et al. (2017b)	Cross-sectional	n = 160 Latino college students from a public college in the East Coast of America	18–46	Loneliness - Revised UCLA Loneliness Scale (R-UCLA) Suicidality - The Beck Hopelessness Scale (BHS)	Correlation analysis between loneliness and suicidal behaviours showed that there was a significant positive correlation ($r = 0.49$, $p \leq 0.001$). In hierarchical regression analysis, with suicidal behaviours as the outcome, step 3

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Table 2 (*continued*)

Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
				- Suicidal Behaviours Questionnaire–Revised (SBQ-R)	showed loneliness as a significant predictor ($\beta = 0.41$, $R^2 = 0.22$, $p \leq 0.001$). Loneliness explained an additional 12 % variance after demographics (age and sex) and ethnic identity were accounted for.
Chang et al. (2017c)	Cross-sectional	$n = 228$ College students from a large public university in the midwest	18–28	Loneliness - Revised UCLA Loneliness Scale (R-UCLA) Suicidality - Beck Depression Inventory (BDI) - Frequency of Suicidal Ideation Inventory (FSII)	Correlational analysis between loneliness and suicidal ideation showed a significant positive association ($r = 0.52$, $p < 0.001$). In hierarchical regression analysis, loneliness was found to account for a 27 % of the variance of suicidal ideation. Future orientation is the notion that one's future can and will improve. The interaction term loneliness × future orientation had a significant negative association, indicating that future orientation moderates the association between loneliness and suicidal ideation. ● Step 1: Loneliness ($\beta = 0.52$, $R^2 = 0.27$, $p < 0.001$) ● Step 2: Future orientation ($\beta = -0.16$, $R^2 = 0.29$, $\Delta R^2 = 02$, $p < 0.05$) ● Step 3: Loneliness × Future orientation ($\beta = -0.99$, $R^2 = 0.35$, $\Delta R^2 = 0.06$, $p < 0.001$).
Chang et al. (2019)	Cross-sectional	$n = 156$ Latino college students from an east coast college	18–46	Loneliness - Revised UCLA Loneliness Scale (R-UCLA) Suicidality - Suicidal Behaviours Questionnaire- Revised (SBQ-R)	A positive significant correlation was exhibited between loneliness and suicidal behaviours ($r = 0.49$, $p < 0.001$). From a hierarchical regression analysis, after accounting for demographics (age and gender), results showed that loneliness was a significant predictor of suicidal behaviours ($\beta = 0.25$, $p < 0.01$). Depression and loneliness accounted for 36 % of the variance in suicidal behaviours, an additional 35 % after demographics were accounted for.
Chesin and Jeglic (2012)	Cross-sectional	$n = 554$ Undergraduate students at an urban public college in New York City	M(SD): 19 (0.14)	Loneliness - UCLA Loneliness Scale-revised (LS) Suicidality - Suicide Behaviours Questionnaire (SBQ)	In multivariate regression analysis, SBQ was predicted from depression, hopelessness, loneliness, social problem solving, ethnic identity, and delinquent behaviour. This model explained 29.5 % of the variance in SBQ scores. ● Loneliness (LS): B (standardized) = 0.08 (0.19), SE.(B) = 0.02, $p < 0.001$
Garcia-Williams et al. (2014)	Cross-sectional	$n = 301$ Graduate students at Emory University between 2010 and 2014	18–63	Loneliness - Negative symptoms (including loneliness) were measured using five items Suicidality - Three items measuring suicidal ideation, planning, and attempt	Students with thoughts of suicide were more likely to feel lonely ($\chi^2(1, N = 300) = 9.14$, $p = 0.002$). Students with plans of suicide were more likely to feel lonely as well ($p = 0.016$).
Joiner Jr and Rudd (1996)	Observational Cohort	$n = 234$ Undergraduate students at a southwestern university, US	18–25	Loneliness - UCLA Loneliness Scale Suicidality - Depressive Symptom Inventory-Suicidality Subscale	T1 (baseline) loneliness predicted increase in suicidal ideation 10 weeks later in T2 (follow-up) (pr = 0.26, $p = 0.026$), but this association was diminished when controlling for T1 hopelessness (pr = 0.02, p = NS). This suggests that the association between loneliness and suicidal ideation depends on the influence of hopelessness.
Keiner et al. (2023)	Cross-sectional	$n = 2016$ Medical students, trainees, and faculty physicians from an anonymous survey	M(SD): 34.30 (11.33)	Loneliness and Suicidality - Stress and Depression Questionnaire from the American Foundation for Suicide Prevention's Interactive Screening Program	Individuals with feelings of loneliness were more likely to report recent suicidal thoughts (UOR = 7.104, $p < 0.001$, $\chi^2 = 173.038$), plans (UOR = 12.958, $p < 0.001$, $\chi^2 = 95.575$), and behaviours (UOR = 6.403, $p < 0.001$, $\chi^2 = 27.913$), as well as having a lifetime history of suicide attempts (UOR = 3.467, $p < 0.001$, $\chi^2 = 21.101$), compared to those who were not lonely.

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Table 2 (*continued*)

Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
Klugman et al. (2024)	Cross-sectional	$n = 337$ Students from different campuses in Israel	≥18	Loneliness - Revised-UCLA Loneliness Scale (R-UCLA) Suicidality - The Suicide Behaviours Questionnaire-Revised (SBQ-R)	A backward stepwise logistic regression analysis was performed. Results revealed that having a sense of loneliness (UOR = 1.047, CI 95 % = 1.014–1.082, $p = 0.005$) is one of the most parsimonious correlates with suicide risk. Loneliness was positively and significantly correlated with suicide risk ($r = 0.326$, $p < 0.001$).
Lamis et al. (2014)	Cross-sectional	$n = 1200$ Undergraduate psychology students from a large southeastern university in the U.S.	18–26	Loneliness - UCLA Loneliness Scale - Version 3 (UCLA-3) Suicidality - Beck Scale for Suicide Ideation (BSS)	Correlational analysis was completed, examining the association between loneliness and suicidal ideation. Controlled covariates include age, gender, race, year in school, sorority/fraternity affiliation, association status, campus residency status, and anxiety sensitivity. ○ Zero-order correlation: ($r = 0.40$, $p < 0.01$) ○ Partial correlation (controlling for covariates): ($r = 0.36$, $p < 0.01$)
Lamlé et al. (2023)	Cross-sectional	$n = 555$ Students attending vocational school for various reasons in Halle/Saale, Germany	≥18	Loneliness - Loneliness measurement (unspecified number of items) Suicidality - Beck depression index (BDI-II) for suicidal ideation	Correlational analysis was calculated between loneliness and suicidal ideation ($r = 0.59$, p = NS). In multivariate linear regression, the overall model included adjusted BDI-score (depressive symptoms), feeling lonely, male sex, importance of religion, satisfied with time for hobbies, BMI outside norm and explained 42 % of the variance in suicidal ideation. Adjusted BDI-score was added first and explained 36.4 % of the variance. Loneliness was added next and adjusted BDI-score and loneliness explained 38.1 % of the variance; loneliness explained an additional 1.7 % of the variance. Loneliness was found to be positively and significantly associated with suicidal ideation $\beta = 0.155$, $p < 0.01$.
Muyan and Chang (2015)	Cross-sectional	$n = 288$ College students attending a public university in Ankara, Turkey	18–40	Loneliness - Revised UCLA Loneliness Scale (R-UCLA) Suicidality - Beck Depression Inventory (BDI) - Frequency of Suicide Ideation Inventory (FSII)	Correlational analysis between loneliness and suicidal ideation showed a significant positive association ($r = 0.29$, $p \leq 0.001$). In hierarchical regression analysis with suicide ideation as the outcome, after step 1 (demographics; age and gender) and step 2 (perfectionism) were accounted for, loneliness was added in step 3. Loneliness was found to increase the likelihood of suicide ideation and add an additional 5 % unique variance ($\beta = 0.23$, $R^2 = 0.19$, $\Delta R^2 = 0.05$, $p \leq 0.001$). In hierarchical regression analysis with suicide ideation as the outcome, after step 1 (demographics; age and gender), step 2 (depressive symptoms), and step 3 (perfectionism) were accounted for, loneliness was added in step 4. Loneliness was found to increase the likelihood of suicide ideation and add an additional 1 % unique variance ($\beta = 0.12$, $R^2 = 0.30$, $\Delta R^2 = 0.01$, $p < 0.05$).
Oh et al. (2024a)	Cross-sectional	$n = 101{,}744$ Data drawn from the 2020–2021 Healthy Minds Study, international and U.S. undergraduate and graduate students	18–29	Loneliness - UCLA Loneliness Scale (UCLA) Suicidality - Suicidal thoughts and behaviours measured using items adapted from the World Health Organization suicide screen	Using multivariable logistic regression models: those who only reported feeling lonely (no cannabis use) had significantly greater odds of suicidal thoughts AOR: 2.75 (no p-value mentioned).
Pitman et al. (2020)	Cross-sectional	$n = 3193$	18–40	Loneliness - Eight items from the Adult	In an unadjusted and adjusted analysis, higher loneliness scores predicted higher

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Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
		Staff and students at 37 UK higher educational institutions with an experience of sudden bereavement of a close friend or relative after the age of 10		Psychiatric Morbidity Surveys Suicidality - Suicide attempt and ideation items were taken from the Adult Psychiatric Morbidity Surveys	probability of post-bereavement suicide attempts and post-bereavement suicidal ideation with significance. Adding scores for perceived stigma of sudden bereavement lowered the associations slightly. In addition, the type of bereavement (from suicide or other causes) did not affect the association between loneliness and suicidality. ● Probability of post-bereavement suicide attempt ○ OR(95 % CI) = 1.22 (1.17–1.27), $p < 0.001$ ○ AOR(95 % CI) = 1.19 (1.14–1.25), $p < 0.001$ ● Addition of perceived stigma of bereavement to adjusted model ○ AOR(95 % CI) = 1.13 (1.08–1.19), $p < 0.001$ ● Probability of post-bereavement suicide ideation ○ OR(95 % CI) = 1.26 (1.22–1.30), $p < 0.001$ ○ AOR(95 % CI) = 1.24 (1.20–1.28), $p < 0.001$ ● Addition of perceived stigma of bereavement to adjusted model ○ AOR(95 % CI) = 1.18 (1.13–1.22), $p < 0.001$
Rich and Bonner (1987)	Cross-sectional	$n = 202$ College Students from the General Psychology Subject at an Eastern University	Estimate: 18–22	Loneliness - UCLA Loneliness Scale Suicidality - Self-Report Scale for Suicide Ideation (SSI) - Self-predicted future suicide probability items from the Suicidal Behaviours Questionnaire (SBQ)	Using pearson's correlation: ● Between loneliness and current suicide ideation ($r = 0.39$, $p < 0.05$). ● Between loneliness and self-predicted future suicide probability ($r = 0.39$, $p < 0.05$). In stepwise multiple regression for predictors of suicidal ideation, depression, few reasons for living, loneliness, and negative life stress were the best predictors where multiple R was 0.55 and $p < 0.05$. 30 % of the variance of suicidal ideation was explained by these variables.
Weber et al. (1997)	Cross-sectional	n = 185 College undergraduate students from a southwestern university	18–25	Loneliness -Revised UCLA Loneliness Scale (R-UCLA) Suicidality -The Suicide Ideation Questionnaire (SIQ)	A significant correlation was found between loneliness and suicidal ideation ($r = 0.52$, $p < 0.01$).
Healthy adolescents (13–17)					
Chan et al. (2016)	Cross-sectional	$n = 2789$ Adolescents from 11 states and 18 districts in Peninsular Malaysia	16–17	Loneliness - Single item measuring loneliness Suicidality - Single item measuring suicidal ideation	In univariate and multivariate analysis, loneliness was significantly and positively associated with suicidal ideation. ● UOR: 6.27 (4.26–9.23), $p < 0.001$ ● AOR: 2.54 (1.57–4.11), $p < 0.001$
Dema et al. (2019)	Cross-sectional	$n = 5809$ Data drawn from the GSHS survey on Bhutan conducted in 2016	13–17	Loneliness - Single item measuring loneliness Suicidality - Single item measuring suicidal ideation - Single item measuring suicidal attempt	Loneliness was associated with suicidal ideation and attempt in logistic regressions. ● Suicidal Ideation ○ PR (95 % CI) = 2.7 (2.6–2.8), $p < 0.001$ ○ aPR (95 % CI) = 1.8 (1.6–2.2), $p < 0.001$ ○ AOR(95 % CI) = 2.1 (1.7–2.7), $p < 0.001$ ● Suicidal Attempt

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Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
					○ PR (95 % CI) = 2.6 (2.4–2.7), p < 0.001 ○ aPR (95 % CI) = 1.4 (1.3–1.7), *p* < 0.001 ○ AOR(95 % CI) = 1.6 (1.2–2.1), p < 0.001
Garnefski et al. (1992)	Cross-sectional	*n* = 570 High school students in the Netherlands, 285 with and 285 without a history of suicidal behaviour	15–16	Loneliness and Suicidality - Monitoring the Future Questionnaire	In bivariate analyses through chi-square, male and female suicide attempters reported higher feelings of loneliness than non-attempters. ● 18 male suicide attempters (n = 100) often felt lonely whereas 11 non-attempters (*n* = 100) often felt lonely (NS) ● 34 female suicide attempters (*n* = 185) often felt lonely whereas 19 non-attempters (n = 185) often felt lonely (NS)
Lagman et al. (2021)	Cross-sectional	*n* = 8761 Data drawn from the GSHS survey, with students in the Philippines from grades 7–10	13–17	Loneliness - Single item measuring loneliness Suicidality - Single item measuring suicide attempt	Through a logistic regression to predict suicide attempts, feeling lonely was a significant predictor (UOR = 2.1, $p \leq 0.0001$, 95 % CI [1.7, 2.5]).
Marthoenis and Nassimbwa (2022)	Cross-sectional	*n* = 4993 Data drawn from Indonesian GSHS of 2015	13–17	Loneliness - Single item measuring loneliness Suicidality - Single item measuring suicide attempts	Odds ratios were calculated for loneliness associated factors, including suicide ideation, plan, and attempt. Adjusted odds ratios adjusted for age, psychosocial distress, health risk behaviour, and protective factors. ● Suicide Ideation ○ UOR (95 % CI) = 4.52 (3.32–6.14), p < 0.001 ○ AOR (95 % CI) = 1.66 (1.01–2.74), p < 0.05 ● Suicide Plan ○ UOR (95 % CI) = 4.39 (3.20–6.03), p < 0.001 ○ AOR (95 % CI) = 2.35 (1.41–3.92), p < 0.001 ● Suicide Attempt ○ UOR (95 % CI) = 3.26 (2.05–5.20), p < 0.001 ○ AOR (95 % CI) = 0.56 (0.28–1.12), p = NS
McKinnon et al. (2016)	Cross-sectional	*n* = 164,770 Data drawn from the WHO-GSHS survey across 32 countries dating from 2003 to 2012	13–17	Loneliness - Single item measuring loneliness Suicidality - Single item measuring suicidal ideation - Single item measuring suicidal planning	Determinants of suicide was consistent across all countries, with loneliness being one of many others having stronger associations. The adjusted risk for suicide was >10 % higher when participants answered with "mostly/always feeling lonely" rather than "never/ rarely feeling lonely." Loneliness was the strongest association to suicide ideation in Region of the Americas, Eastern Mediterranean and South-East Asia and Western Pacific Regions. Substantial heterogeneity was found in the association between loneliness and suicidal ideation (heterogeneity *p*-values <0.001).
Peltzer and Pengpid (2017)	Cross-sectional	n = 30,284 Data drawn from the GSHS from seven ASEAN countries (Cambodia, Indonesia, Malaysia, Myanmar, Philippines, Thailand and Vietnam)	13–15	Loneliness - Single item measuring loneliness Suicidality - Single item measuring suicidal ideation	In multivariable logistic regression analysis, loneliness was positively and significantly associated with suicidal ideation prevalence in the total sample and when the sample was separated into boys and girls. ● All: AOR (95 % CI) = 3.19 (2.47–4.11), *p* < 0.001 ● Boys: AOR (95 % CI) = 3.61 (2.42–5.38), p < 0.001

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Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
					● Girls: AOR (95 % CI) = 2.99 (2.19–4.07), p < 0.001
Sarfo et al. (2023)	Cross-sectional	$n = 1779$ Data drawn from the 2017 GSHS on school-aged adolescents in Samoa	13–17	Loneliness - Single item question measuring loneliness Suicidality - Three items measuring suicidal ideation, planning, and attempt	In logistic regression, loneliness was positively associated with suicidal behaviours, with suicidal ideation and attempt being significant. ● Suicidal Ideation: AOR (95%CI) = 1.60 (1.125–2.286), $p < 0.01$ ● Suicidal Plan: AOR (95%CI) = 1.40 (0.973–2.007), p = NS ● Suicidal Attempt: AOR (95%CI) = 1.88 (1.303–2.720), p < 0.01
Shirima et al. (2023)	Cross-sectional	$n = 4188$ (3182 in 2019 Survey and 1006 in 2022 Survey) Data drawn from two separate studies in 2019 and 2022 years from the RSHS in the Kilimanjaro region, northern Tanzania on adolescents attending public secondary schools	13–17	Loneliness - Single item measuring loneliness Suicidality - Single item measuring suicide attempt - Two items measuring suicidal ideation and planning	In adjusted analysis, loneliness was found to lead to higher odds of suicidal attempts. This finding was significant in both studies combined, but insignificant in the studies separately. ● Overall (both studies) ○ AOR (95 % CI) = 2.2 (1.1, 4.3), $p = 0.020$ ● Study 1 ○ AOR (95 % CI) = 2.4 (1.0, 5.7), $p = 0.055$ ● Study 2 ○ AOR (95 % CI) = 1.9 (0.7, 5.5), $p = 0.215$
Healthy adults and older adults (≥18)					
Bennardi et al. (2017)	Observational Cohort	$n = 2392$ Non-institutionalized adults from the Spanish population	18–98	Loneliness - UCLA Loneliness Scale Suicidality - WHO Composite International Diagnostic Interview	Odds ratios were calculated between suicide ideation and loneliness. A positive and significant association was discovered in the 60 years or older group. This was a longitudinal study, where follow-up data collection occurred 12 months after. ● Total Sample ○ UOR(95 % CI) = 1.01 (0.99, 1.02), p = NS ● 18–59 Years ○ UOR(95 % CI) = 1.00 (0.98, 1.02), p = NS ● 60+ Years ○ UOR(95 % CI) = 1.02 (1.01, 1.04), $p = 0.009$
Ernst et al. (2021)	Cross-sectional	$n = 2450$ Data drawn in 2014, directed by the University of Leipzig	M(SD): 50.51 (17.03)	Loneliness - R-UCLA Loneliness Scale Suicidality - German Beck Scale for Suicide Ideation (BSS)	Correlational analysis between loneliness and suicidal ideation showed that there was a significant positive correlation ($r = 0.275$, $p \leq 0.01$). Correlational analysis stratified by gender showed a difference in strength of the correlation between loneliness and suicidal ideation.There was a stronger correlation in men. ● Women: $r = 0.247$, $p \leq 0.01$ ● Men: $r = 0.309$, $p \leq 0.01$ The linear regression analysis model included age, gender, equivalized income, living with others, mental distress, loneliness, loneliness × gender, loneliness × age, gender × age, and loneliness × gender × age as predictors of suicidal ideation. The model overall explained 13 % of the variance in suicidal ideation. ● Loneliness: B(95 % CI) = 0.036 (0.0106; 0.149), $\beta = 0.410$, $p < 0.001$

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Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
					● Loneliness × Gender: B(95 % CI) = −0.036 (−0.067; −0.005), β = −0.281, p = 0.024 ● Loneliness × Age: B(95 % CI) = −0.010 (−0.020;-0.001), β = −0.255, p = 0.033 ● Loneliness × Gender × Age: B(95 % CI) = 0.012 (−0.007;0.032), β = 0.180, p = NS This suggests that loneliness and suicidal ideation is moderated by gender and age. Specifically, the association is stronger in one gender than the other, and the strength of the association decreases with age. However, the moderating effect of gender and age had a non-significant difference. The linear regression analysis with the same predictors was stratified by gender. In the women group, the model predicted 9.4 % of the variance in suicidal ideation, while in the men group, 17.8 % of the variance was explained by the predictors, revealing a stronger association from men. Additionally, it was found that the effect of loneliness on suicidal ideation decreased with age. ● Women ○ Loneliness: B(95 % CI B) = 0.046 (0.004; 0.088), β = 0.113, p = 0.035 ○ Loneliness × Age: B(95 % CI B) = −0.004 (−0.017; 0.009), β = −0.033, p = 0.58 ● Men ○ Loneliness: B(95 % CI B) = 0.078 (0.032; 0.125), β = 0.192, p = 0.001 ○ Loneliness × Age: B(95 % CI B) = 0.018 (−0.032; −0.004), β = −0.149, p = 0.014
Gomboc et al. (2022)	Cross-sectional	n = 991 Data drawn from the Mental Health Literacy Research Project in Slovenia from 2018 to 2020	18–95	Loneliness - De Jong Gierveld Loneliness Scale Suicidality - Paykel Suicide Scale	After block one of a hierarchical regression analysis, emotional and social loneliness were added and assessed as predictors of suicidal ideation in block 2. These predictors explained an additional 9 % of variance after accounting for gender, civil status, education, employment status, personal income, suicide attempt, stress, and low well-being in block one. ● Block 2: R^{2a} = 0.29, χ^2 = 168.62, Df = 14, p < 0.001 ● Emotional Loneliness: UOR = 1.86, p < 0.01 ● Social Loneliness: UOR = 1.37, p < 0.004 In logistic regression analysis that separated different age groups, odds ratios were recorded for emotional and social loneliness as predictors of suicidal ideation. ● Emotional Loneliness ○ 18–29 years: UOR (95 % CI) = 1.93 (1.26, 2.95), p = 0.02 ○ 30–44 years: UOR (95 % CI) = 1.60 (1.09, 2.36), p = NS ○ 45–64 years: UOR (95 % CI) = 2.11 (1.42, 3.12), p < 0.001 ○ 65+ years: UOR (95 % CI) = 2.68 (1.40, 5.14), p = 0.003 ● Social Loneliness ○ 18–29 years: UOR (95 % CI) = 1.44 (0.98, 2.11), p = NS ○ 30–44 years: UOR (95 % CI) = 2.42 (1.31, 4.46), p = 0.005 ○ 45–64 years: UOR (95 % CI) = 0.88 (0.64, 1.20), p = NS

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Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
					○ 65+ years: UOR (95 % CI) = 2.62 (1.25, 5.46), $p = 0.011$
Kim et al. (2022)	Cross-sectional	$n = 3000$ 1500 each year Data drawn from combining the 2013 and 2018 South Korean National Suicide Survey	19–75	Loneliness and Suicidality - Questionnaire on Attitudes Towards Suicide (ATTS)	Through factor analysis, loneliness explained 5.32 % of the variance in ATTS scores where "loneliness that drives people to suicide" had a coefficient of 0.777 and "loneliness can be a reason for suicide" had a coefficient of 0.640. When comparing loneliness with different categories along the suicide continuum, the mean score of loneliness attitude towards suicide increased. ● No suicidal idea: M(SD): 3.26(0.87) ● Suicidal idea only: M(SD): 3.38(0.85) ● Suicidal plan: M(SD): 3.35(0.82) ● Suicidal attempt: M(SD): 3.70(0.82) In the univariate and multivariate model loneliness was associated with increased odds of suicidal idea, plan, and suicide attempt. The multivariate model adjusted for age, sex, marital status, education, income and religion. There was significance between loneliness and suicidal idea and suicide attempt in the univariate model and loneliness and suicidal attempt in the multivariate model. ● Univariate Model ○ Suicidal Idea: UOR (95 % CI) = 1.19 (1.06–1.33), $p = 0.004$ ○ Suicidal Plan: UOR (95 % CI) = 1.14 (0.89–1.47), $p = 0.307$ ○ Suicide Attempt: UOR (95 % CI) = 1.93 (1.37–2.71), $p < 0.001$ ● Multivariate Model ○ Suicidal Idea: AOR (95 % CI) = 1.08 (0.95–1.22), $p = 0.259$ ○ Suicidal Plan: AOR (95 % CI) =0.91 (0.68–1.21), $p = 0.501$ ○ Suicide Attempt: AOR (95 % CI) = 1.50 (1.03–2.20), $p = 0.036$
Larson et al. (2024)	Cross-sectional	$n = 1134$ Data drawn from the Arizona Veteran Survey; female veterans and service members	≥18	Loneliness -Single item measuring loneliness Suicidality -Three items measuring suicidality	Using chi-squared analysis, a significant association was found with experiencing loneliness or isolation and suicidality in the last two weeks (1, $n = 992$) = 67.40, $p < 0.001$.
Li et al. (2016)	Cross-sectional	n = 15,957 Data drawn from Sample Survey on Aged Population in Urban/Rural China directed in 2006 by the CRCA	60–103	Loneliness - Single item measuring loneliness Suicidality - Three items measuring suicidal ideation - Three items measuring suicide attempt	In logistic regression analysis on suicidal thoughts and suidice attempts, loneliness increased the likelihood of both in the previous month, previous 12 months, and previous 5 years. The only exception was that for suicide attempts in the previous month, loneliness was not a significant predictor. Suicidal Thoughts and Loneliness ● Previous month: UOR (95 % CI) = 1.70 (1.01, 2.86), $p < 0.008$ ● Previous 12 months: UOR (95 % CI) = 2.41 (1.61, 3.57), $p < 0.008$ ● Previous 5 years: UOR (95 % CI) = 2.11 (1.57, 2.81), $p < 0.008$ Suicide Attempts and Loneliness ● Previous month: UOR (95 % CI) = 1.71 (0.79, 3.73), $p > 0.008$ ● Previous 12 months: UOR (95 % CI) = 1.05 (1.54, 6.97), $p < 0.008$

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Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
					● Previous 5 years: UOR (95 % CI) = 3.13 (1.88, 5.22), p < 0.008
McClelland et al. (2023)	Cross-sectional	$n = 582$ Participants were gathered through social media advertisements at the University of Glasgow	18–70	Loneliness - UCLA version 3 - Social and Emotional Loneliness Scale for Adults (SELSA) Suicidality - The Suicide Probability Scale (SPS)	Depression (depressive symptoms) was tested as a mediator of the association between SELSA-Family, SELSA-Romantic, SELSA-Social, and UCLA and suicidal ideation. ● The indirect effect of depression on the association between SELSA-Family and suicidal ideation was significant (β = 0.044, 95 % CI: 0.014, 0.074), suggesting that depression partially mediated the association. ● The effect of depression on the association between SELSA-Romantic and suicidal ideation reduced the association to non-significant (β = 0.011, p = 0.264, 95 % CI: 0.008, 0.030), suggesting that depression fully mediated the association. ● The indirect effect of depression on the association between SELSA-Social and suicidal ideation was non-significant (β = −0.024, 95 % CI: 0.076, 0.024), suggesting that depression did not mediate the association. ● The indirect effect of depression on the association between UCLA-LS and suicidal ideation was significant (β = 0.126, 95 % CI: 0.105, 0.150), suggesting that depression partially mediated the association.
Ofei-Dodoo et al. (2020)	Cross-sectional	$n = 197$ Physicians of the Medical Society of Sedgwick County	≥25	Loneliness - UCLA Loneliness Scale (UCLA) Suicidality - Single item question measuring suicidal ideation	No significant association was exhibited between loneliness and suicidal ideation (UOR (95 % CI) = 0.17 (0.02, 1.62), p = 0.23).
Vuscan et al. (2022)	Cross-sectional	$n = 1102$ Participants of the non-institutionalized adult and older adult population living in Romania.	≥18	Loneliness - The Short Scale of Socio-Emotional Loneliness for perceived loneliness (social loneliness and emotional loneliness) Suicidality - Single item measuring suicide ideation	Significant correlations between suicidal ideation and items of the socio-emotional loneliness scale: ● 1 (I experience a general sense of emptiness): $r = 0.271$, $p \leq 0.01$ ● 2 (I miss having people around): $r = -0.017$, p = NS ● 3 (I often feel rejected): $r = 0.254$, p ≤ 0.01 ● 4 (There are plenty of people I can lean on in case of trouble): $r = -0.007$, p = NS ● 5 (There are many people I can count on completely): $r = 0.097$, p ≤ 0.01 ● 6 (There are enough people that I feel close to): $r = 0.080$, p ≤ 0.01 ● 7 (Total socio-emotional loneliness score): $r = 0.203$, p ≤ 0.01
Zhang et al. (2021)	Cross-sectional	$n = 538$ Nursing home residents from 7 districts of China	≥60	Loneliness - UCLA Loneliness Scale Suicidality - Beck Scale for Suicidal Ideation-Chinese Version-Current (BSI-CV-C)	Bivariate correlation analysis showed a significant and positive correlation between loneliness and suicidal ideation ($r = 0.375$, p < 0.001). Loneliness has been found to have a positive effect on suicidal ideation, even when resilience acts as a mediator: ● Loneliness: B = 0.257, LLCI = 0.165, ULCI = 0.349, p < 0.001 ● Overall model: $R^2_{adj} = 0.351$ In a model examining the moderating effect of social support from family, friends, and nursing staff on the association between

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Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
					loneliness and suicidal ideation, loneliness was positively and significantly associated with suicidal ideation, but moderated by family support. Family support significantly decreased the strength of the association, but friends support and nursing staff support did not significantly moderate the association. ● Moderator: Family Support ○ Loneliness: β (95 % CI): 0.261(0.166, 0.356), p < 0.001 ○ Loneliness × family support: β (95 % CI): −0.111 (−0.195, −0.027), p < 0.001 ● Moderator: Friends Support ○ Loneliness: β (95 % CI): 0.249 (0.152, 0.346), p < 0.001 ○ Loneliness × friends support: β (95 % CI): −0.022 (−0.107, 0.063), p = NS ● Moderator: Nursing Staff Support ○ Loneliness: β (95 % CI): 0.205 (0.110, 0.300), *p* < 0.001 ○ Loneliness × staff support: β (95 % CI): 0.019 (−0.061, 0.099), p = NS
Healthy adults and adolescents (13–64)					
Cleverley and Kidd (2011)	Cross-sectional	*n* = 47 Homeless youth from youth-serving agencies in Hamilton, Ontario, Canada	15–21	Loneliness - UCLA Loneliness Scale Suicidality - 4-item scale for suicidal ideation	Psychosocial distress was measured using 5 subscales, one including the UCLA Loneliness Scale. Correlational analysis showed that suicidal ideation was positively associated with psychosocial distress (*r* = 0.58). Significance was not recorded. In univariate regression, psychodistress significantly and positively predicted suicidal ideation scores (B = 0.157, S.E. = 0.033, β = 0.575, *t* = 4.718, *p* < 0.001).
Kirkpatrick-Smith et al. (1992)	Cross-sectional	*n* = 613 Students from an upper middle-class, suburban school district in western Pennsylvania	14–19	Loneliness - UCLA Loneliness Scale Suicidality - Self-report version of the Scale for Suicidal Ideation (SSI)	Suicide ideation was positively and significantly associated with loneliness (*r* = 0.32, *p* < 0.05).
Lasgaard et al. (2011a)	Cross-sectional	*n* = 1009 Data drawn from the High School Loneliness Study, a Danish national study	15–26	Loneliness - Social and Emotional Loneliness Scale for Adults (SELSA-S) Suicidality - 8-item Suicide Ideation subscale from the Suicide Probability Scale (SPS-SI)	Significant correlation was found between peer-related loneliness and suicide ideation (r = 0.27, *p* < 0.006), family-related loneliness and suicide ideation (*r* = 0.34, p < 0.006), and romantic loneliness and suicide ideation (*r* = 0.10, p < 0.006). In hierarchical regression analysis, in step 2 with suicide ideation as the outcome, peer-related loneliness (β = 0.17, r^2_{sp} = 0.027), family-related loneliness (β = 0.26, r^2_{sp} = 0.061), and romantic loneliness (β = 0.04, r^2_{sp} = 0.002) were significant predictors. The variables in step 2 explained 14 % additional variance in suicide ideation after step one controlling for covariates including gender, age, living in a two-parent family, living with boyfriend/girlfriend, and living alone, (ΔR^2 (%) = 14).
Lasgaard et al. (2011b)	Cross-sectional and observational cohort	n = 1009 Data drawn from the High School Loneliness Study, a Danish national study	15–26	Loneliness - Social and Emotional Loneliness Scale for Adults (SELSA-S) Suicidality - 8-item Suicide Ideation subscale from the Suicide Probability Scale (SPS-SI)	Cross-Sectional Study Significant correlation was found between loneliness and suicide ideation (r = 0.39, *p* < 0.0005). In hierarchical regression, the association between suicide ideation and multiple predictors was assessed.

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Table 2 (*continued*)

Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
					● Step 1 only included loneliness and yielded $\beta = 0.38$, $p < 0.0005$ and remained consistent in step 2 after age, gender, parental education, two-parent family, living with boyfriend/girlfriend, living alone, residential location, birth place, boyfriend/girlfriend, and high school course variables were added. ● After step 3 added depressive symptoms, loneliness dropped to $\beta = -0.05$, p = NS and remained consistent after social desirability was added in step 4, indicating that loneliness's effect on suicide ideation was reduced and no longer significant after depressive symptoms were taken to account. ● Depressive symptoms may be a key mediator between loneliness and suicide ideation. ● Loneliness by itself in step 1 explained 14 % of the variance in suicide ideation. $R^2(\%) = 14$, F (1947) = 159 Longitudinal Study ● T1 is baseline, T2 is one year after. ● Loneliness(T1) and Suicide Ideation(T1): $r = 0.25$, $p < 0.0005$ ● Loneliness(T1) and Suicide Ideation(T2): $r = 0.20$, $p < 0.0005$ ● Loneliness(T2) and Suicide Ideation(T1): $r = 0.25$, $p < 0.0005$ ● Loneliness(T2) and Suicide Ideation(T2): $r = 0.37$, $p < 0.0005$ ● Increased loneliness was correlated with increased suicide ideation at different time points.
Page et al. (2006)	Cross-sectional	$n = 8450$ (2624 from Taiwan, 2519 Thailand, 3307 Philippines) High school students from Taiwan, Thailand, and the Philippines	Estimate: 15–18	Loneliness - Revised UCLA Loneliness Scale (R-UCLA) Suicidality - Two items measuring suicide attempts	ANCOVA tested the difference between suicide attempter and non-attempters on loneliness. For boys: ● Ever attempted suicide ○ Taiwan students ■ Attempters: M(SE) = 44.56 (0.66) ■ Non-Attempters: M(SE) = 39.79 (0.29) ■ F(df) = 43.14 (3, 1550), $p < 0.0001$ ○ Thailand students ■ Attempters: M(SE) = 43.99 (1.36) ■ Non-Attempters: M(SE) = 38.38 (0.31) ■ F(df) = 16.02 (3, 750), $p < 0.0001$ ○ Philippine students ■ Attempters: M(SE) = 43.99 (1.36) ■ Non-Attempters: M(SE) = 38.38 (0.31) ■ F(df) = 16.02 (3, 750), $p < 0.0001$ ● Attempted suicide in the past 12 months ○ Taiwan students ■ Attempters: M(SE) = 43.76 (0.75) ■ Non-Attempters: M(SE) = 41.18 (0.26) ■ F(df) = 10.52 (3, 956), $p = 0.0012$ ○ Thailand students ■ Attempters: M(SE) = 47.29 (1.04) ■ Non-Attempters: M(SE) = 40.13 (0.28) ■ F(df) = 43.86 (3, 1547), $p < 0.0001$ ○ Philippine students ■ Attempters: M(SE) = 44.11 (1.89) ■ Non-Attempters: M(SE) = 38.52 (0.31) ■ F(df) = 8.47 (3, 750), $p = 0.0037$

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Table 2 (*continued*)

Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
					For girls: ● Ever attempted suicide ○ Taiwan students ■ Attempters: M(SE) = 42.07 (0.58) ■ Non-Attempters: M(SE) = 37.70 (0.37) ■ F(df) = 39.97 (3,1022), p < 0.0001 ○ Thailand students ■ Attempters: M(SE) = 40.22 (0.73) ■ Non-Attempters: M(SE) = 36.11 (0.20) ■ F(df) = 29.43 (3, 1560), p < 0.0001 ○ Philippine students ■ Attempters: M(SE) = 43.61 (0.63) ■ Non-Attempters: M(SE) = 40.59 (0.21) ■ F(df) = 20.45 (3, 1393), p < 0.0001 ● Attempted suicide in the past 12 months ○ Taiwan students ■ Attempters: M(SE) = 44.14 (0.37) ■ Non-Attempters: M(SE) = 38.17 (0.33) ■ F(df) = 40.15 (3, 1017), p < 0.0001 ○ Thailand students ■ Attempters: M(SE) = 40.17 (0.87) ■ Non-Attempters: M(SE) = 36.23 (0.20) ■ F(df) = 19.32 (3, 1558), p < 0.0001 ○ Philippine students ■ Attempters: M(SE) = 43.55 (0.66) ■ Non-Attempters: M(SE) = 40.62 (0.21) ■ F(df) = 17.62 (3, 1400), p < 0.0001
Pereira and Cardoso (2018)	Cross-sectional	*n* = 366 Students from a university in northern Portugal	Females M(SD): 20.86 (3.75) Males M(SD): 21.65 (4.45)	Loneliness - UCLA Loneliness Scale Suicidality - Suicidal Ideation Questionnaire (SIQ)	Males with significant suicidal ideation compared with other males with non-signficant suicidal ideation scored significantly higher on levels of loneliness (U = 234, p < 0.05, δ = 0.619). Females with significant suicidal ideation compared with other females with non-signficant suicidal ideation scored significantly higher on levels of loneliness (U = 1178.5, p < 0.05, δ = 0.602). A regression model was developed to predict suicidal ideation based on psychological variables, including depressive symptomathology, trust in others, comfort with intimacy, anxiety, social anxiety, and loneliness (UCLA Loneliness Scale). This model effectively differentiated between non-suicidal ideation (NSI) and suicidal ideation (SSI) groups (chi-square = 79.351, *p* < 0.001, df = 6), accounting for 39.5 % of the variance in suicidal ideation. ● Loneliness: B = 0.091, UOR = 1.095, *p* < 0.01, Exp(B) 95 % CI = [1.021, 1.175]
Putra et al. (2019)	Cross-sectional	*n* = 8634 (5003 female, 3631 male) Data drawn from the GSHS of Indonesia in 2015	13–18	Loneliness - Single item measuring loneliness Suicidality - Single item measuring suicidal ideation - Single item measuring suicide attempt	Odd ratios were calculated for suicidal ideation and attempt when loneliness is present as a factor. This was calculated for the total sample and separately for male and female. ● Suicidal Ideation ○ Total: UOR (95 % CI) = 2.69 (1.84–3.92), p < 0.001 ○ Female: UOR (95 % CI) = 2.11 (1.25–3.57), p < 0.01 ○ Male: UOR (95 % CI) = 4.37 (2.36–8.11), p < 0.001 ● Suicide Attempt

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Table 2 (*continued*)

Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
					○ Total: UOR (95 % CI) = 1.86 (1.13–3.07), p < 0.05 ○ Female: UOR (95 % CI) = 1.4 (0.79–2.55), p = NS ○ Male: UOR (95 % CI) = 2.82 (1.08–7.36), p < 0.05
Yadegarfard et al. (2014)	Cross-sectional	*n* = 260 (129 transgender and 131 cisgender male) Transgender participants were from an association based in Bangkok: the Rainbow Sky Association Cisgender male students were from the Assumption University of Thailand	15–25	Loneliness - UCLA Loneliness Suicidality - Suicidal tendency (thoughts and attempts): Positive and Negative Suicide Ideation Inventory (PANSI)	In multiple regression analysis of transgendered respondents, the higher the reported loneliness the higher the frequency of negative risk factors of suicidal tendency (PANSI-Negative, β = 0.45) and protective factors of suicidal tendency (PANSI-Positive, β = 0.24). These findings were significant. In multiple regression analysis of cisgender respondents, the higher the reported loneliness the higher the frequency of negative risk factors of suicidal tendency (PANSI-Negative, β = 0.25) and the lower the frequency of protective factors of suicidal tendency (PANSI-Positive, β = −0.34). These findings were significant.
Yang et al. (2023a)	Observational Cohort	*n* = 3267 High school students from six counties or districts in Taizhou City, China	Estimate: 14–18	Loneliness - UCLA Loneliness Scale Suicidality - One item from the Chinese version of the Children's Depression Inventory (CDI)	Three models were used to evaluate loneliness predicting suicidal ideation. Model 1 did not control for covariates, model 2 adjusted for sociodemographic covariates and baseline suicidal ideation, and model 3 adjusted for baseline depressive symptoms. There was a positive and significant association in model 1 and 2, but model 3 did not have a significant association. Model 1 ● UOR (95 % CI) = 1.88 (1.21–2.92), *p* = 0.005 Model 2 ● UOR (95 % CI) = 1.89 (1.20–2.97), *p* = 0.006 Model 3 ● UOR (95 % CI) = 1.09 (0.65–1.82), *p* = 0.750
Zhou et al. (2023)	Cross-sectional	*n* = 393 Undergraduate students from a university in Chengdu, China	16–25	Loneliness - UCLA Loneliness Scale Suicidality - Suicidal Behaviours Questionnaire-Revised (SBQ-R)	In bivariate correlation analysis, loneliness was positively and significantly associated with suicidal behaviour (*r* = 0.440, p < 0.01). In hierarchical analysis, loneliness explained an additional unique 19 % variance of suicidal behaviour after demographics (age and sex) were accounted for. Loneliness was positively and significantly associated with suicidal behaviour, but this association was diminished in the loneliness × school belonging addition due to school belonging's mediation effect. ● Step 2 Loneliness ○ β = 0.44, R^2 = 0.21, ΔR^2 = 0.19, p < 0.001 ● Step 3 School Belonging ○ β = −0.25, R^2 = 0.24, ΔR^2 = 0.03, p < 0.001 ● Step 4 Loneliness × School Belonging ○ β = −0.51, R^2 = 0.25, ΔR^2 = 0.02, p < 0.001
Persons with major depressive disorder (MDD)					
Teo et al. (2018)	Observational Cohort	n = 301	M(SD): 55.4 (14)	Loneliness - NIH Toolbox Adult Social	In multivariate models that accounted for sociodemographic confounders, loneliness

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Table 2 (*continued*)

Author(s)	Study type	Sample population and size (n =)	Age range	Assessment tools	Outcomes of interest
		Eligible veteran participants invited to the study through Veterans Health Administration's Corporate Data Warehouse Participants were highly likely to have major depression (PHQ-8 score ≥ 9 at time of telephone screening) but were not formally diagnosed with MDD		Association Scales Suicidality - Depressive Symptom Index Suicidality Subscale (DSI-SS)	was linked to increased levels of depression and suicidal ideation.
Persons with bipolar disorder (BD)					
Pike (2024)	Case-Control	n = 146 (86 BD and 60 healthy controls) Participants are from the Mood and Psychosis Research Program at Brigham and Women's Hospital, Boston Participants had BD1 or BD2	18–68	Loneliness - Revised Loneliness Scale (R-UCLA) Suicidality - Number of Lifetime Suicide Attempts using SCID-5	In a logistic regression of BD patients controlling for depression severity, those with lifetime suicide attempt status experienced higher R-UCLA scores of loneliness than those without. (n = 84, B = 0.04, SE = 0.02, UOR = 1.04, CI = 1.00–1.08, $p = 0.056$). In a linear regression of BD patients controlling for depression severity, an increase in number of lifetime suicide attempts correlated with higher R-UCLA scores of loneliness. ($n = 84$, B = 0.01, SE = 0.02, β = 0.10, $r = 0.09$, $p = 0.412$). In a logistic regression of BD patients controlling for depression severity, those with current suicidal ideation experienced higher R-UCLA scores of loneliness than those without. ($n = 86$, B = 0.08, SE = 0.04, UOR = 1.08, CI = 1.01–1.16, $p = 0.033$).
Persons with persistent depressive disorder (PDD)					
Wolf et al. (2022)	Prospective Naturalistic Intervention	$n = 52$ (26 PDD, 26 BPD clinical controls) Data drawn from participants in the prospective naturalistic study at the Department of Psychiatry and Psychotherapy of the LMU University Hospital, Munich	PDD - M(SD): 36.8(12.1) BPD - M(SD): 26.6(8.5)	Loneliness - Loneliness Scale (UCLA) Suicidality - Columbia-Suicide Severity Rating Scale (C-SSRS) to measure SIB (any suicidal ideation or behaviour on admission)	In a correlational analysis of 26 PDD patients before CBASP treatment, there was a significant correlation between UCLA and suicidal ideation and behaviours. ● UCLA Loneliness ○ Severity of Ideation: r = 0.32, p = NS ○ Intensity of Ideation: $r = -0.06$, p = NS ○ Severity of Behaviour: $r = 0.40$, $p \leq 0.05$ ● UCLA Feelings of Loneliness ○ Severity of Ideation: $r = 0.24$, p = NS ○ Intensity of Ideation: $r = 0.00$, p = NS ○ Severity of Behaviour: $r = 0.15$, p = NS
Persons with various depressive and bipolar disorders					
Wiktorsson et al. (2010)	Case-Control	$n = 511$ (103 with affective disorders, 408 comparison group) Patients admitted to emergency ward (2003–2006) from five hospitals in western Sweden for suicide attempt Comparison group from Gerontological and Geriatric Population Studies and the Prospective Population Study of Women, Gothenburg	≥70	Loneliness - Single item measuring loneliness	Participants had either MDD, unipolar depression, BD, minor depression or other depressive disorders. However, the comparison group did not have BP patients. Participants also had dementia or not. Comparison group matched cases in sex and age group (70–73, 74–77, 78–81, 82–85, and 86–91 years) 60 % of non-demented suicide attempters reported loneliness compared to 17.6 % in the non-demented comparison group (UOR: 7.07, 95 % CI: 4.2–12.0, $p = 0.000$), remaining significant after controlling for depression (UOR: 2.8, 95 % CI: 1.3–6.1, $p = 0.010$).

M(SD) = Mean(Standard Deviation); M(SE) = Mean(Standard Error); MDD = Major Depressive Disorder; BD = Bipolar Disorder; BD1 = Bipolar 1 Disorder, BD2 = Bipolar 2 Disorder; PDD = Persistent Depressive Disorder; GSHS = Global School-Based Student Health Survey; ASEAN = Association of Southeast Asian Nations; RSHS = Region School Health Survey; UCLA-LS = University of California Los Angeles Loneliness Scale; R-UCLA = Revised UCLA Loneliness Sclae; UCLA-3 = UCLA Loneliness Scale - Version 3; SELSA = Social and Emotional Loneliness Scale for Adults; SELSA-S = SELSA-Shorter Version; SELSA-Family = SELSA-Family subscale; SELSA-Romantic = SELSA-Romantic subscale; SELSA-Social = SELSA-Social subscale; ATTS = Questionnaire on Attitudes Towards Suicide; SPS = Suicide Probability Scale; SPS-SI = Suicide Probability Scale - Suicide Ideation; SSI = Scale for Sucidal Ideation; SBQ = Suicidal Behaviours Questionnaire; SBQ-R = SBQ-Revised; FSII = Frequency of Suicidal Ideation Inventory; BHS = Beck Hopelessness Scale; BDI-II = Beck Depression Index; BSS = Beck Scale for Suicide Ideation; PANSI = Positive and Negative Suicide Ideation Inventory; PANSI-Positive = PANSI Positive Suicide Ideation subscale; PANSI-Negative = PANSI Positive Suicide Ideation subscale;

Children's Depression Inventory (CDI); DSI-SS = Depressive Symptom Index Suicidality Subscale; C-SSRS = Columbia-Suicide Severity Rating Scale; SIB = Suicidal Ideation or Behaviour; LECCS-NEG = Life Events Checklist for College Students; CBASP = Cognitive Behavioural Analysis System of Psychotherapy; NS = Non-Significant; UOR = unadjusted odd ratio, AOR = adjusted odd ratio, T1 = baseline, T2 = follow-up.

2017a, 2017b, 2017c, 2019; Chesin and Jeglic, 2012; Garcia-Williams et al., 2014; Joiner Jr and Rudd, 1996; Keiner et al., 2023; Klugman et al., 2024; Lamis et al., 2014; Lamlé et al., 2023; Muyan and Chang, 2015; Oh et al., 2024a; Pitman et al., 2020; Rich and Bonner, 1987; Weber et al., 1997). A consideration for these findings is that college and university students were the most common sample group across studies, reflecting a focus on emerging adults and academic settings.

In studies that included both healthy adults and adolescents (13 to 64 years old), correlations ranged from $r = 0.10$ to 0.58 (Cleverley and Kidd, 2011; Kirkpatrick-Smith et al., 1992; Lasgaard et al., 2011a, b; Page et al., 2006; Pereira and Cardoso, 2018; Putra et al., 2019; Yadegarfard et al., 2014; Yang et al., 2023a; Zhou et al., 2023). While the upper end of this range align with the findings in healthy adult-only studies, the lower end suggested a weaker or less consistent association among healthy adolescents. Overall, the variation in the strength of positive correlations indicates that age was a moderating factor, with a moderate association observed in both healthy adults and older adults, and a potentially weaker, though positive, association in adolescents. It is also important to note that college and university students were the most frequently studied sample population in studies that included healthy adults and adolescents, reflecting an emphasis on young adults.

Studies with healthy adolescents (13 to 17 years old) were also mostly cross-sectional and rated as fair in risk of bias assessments (Chan et al., 2016; Dema et al., 2019; Garnefski et al., 1992; Lagman et al., 2021; Marthoenis and Nassimbwa, 2022; McKinnon et al., 2016; Peltzer and Pengpid, 2017; Sarfo et al., 2023; Shirima et al., 2023). These studies found a moderate to strong positive association, UOR = 2.1 to 6.27. After adjustment, the associations ranged from weak negative to moderate positive, AOR = 0.56 to 2.4. In comparison, for healthy adults, UORs showed a weak to strong positive association, UOR = 1.047 to 12.958, with adjusted results showing a weak to moderate positive association, AOR = 1.18 to 2.75. The weak negative association in adolescents was inconsistent with the positive correlation found previously. In addition, the larger variability in UORs in adults compared to adolescents suggested that covariates present in both age groups may have been more influential in adults. Notwithstanding these findings, variability in these odds ratios may have been due to different sample characteristics, measures used, and other heterogeneity in study design across studies.

Studies with healthy adults and older adults (≥18 years old) found a strong negative to positive association, UOR = 0.17 to 3.13 (Bennardi et al., 2017; Ernst et al., 2021; Gomboc et al., 2022; Kim et al., 2022; Larson et al., 2024; Li et al., 2016; McClelland et al., 2023; Ofei-Dodoo et al., 2020; Vuscan et al., 2022; Zhang et al., 2021). After adjustment, a weak negative to positive association was found, AOR = 0.91 to 1.50. Comparing AORs between healthy adults and older adults and adults-only, a weak negative association was found in adults and older adults studies that did not exist in adults-only studies. This indicated that there might be a negative association in older adults, which is inconsistent with the positive correlation previously found. In contrast, the positive association found in adults was consistent with the positive correlation. All three age groups demonstrated an association between loneliness and suicidality, however.

3.4. Depressive or bipolar disorder patients

Samples of patients with depressive or bipolar disorders were gathered from the United States, Germany, and Europe. Various loneliness and suicidality scales were employed, including UCLA and NIH Toolbox Adult Social Association Scales for loneliness and Depressive Symptom Index Suicidality Subscale (DSI-SS), SCID-5, and Columbia-Suicide Severity Rating Scale (C-SSRS) for suicidality. Samples consisted mostly of adults to older adults.

A study on enrolled patients with MDD did not report statistics, but generally found that loneliness was connected to increased levels of depression and suicidal ideation (Teo et al., 2018). This was an observational study with 301 participants and was rated fair in quality assessment.

In the study on enrolled BD patients, a weak positive correlation was recorded, $r = 0.09$, and a weak positive association was found, UOR = 1.04 to 1.08 (Pike et al., 2024). This was a case-control study with 146 participants, of which 86 had BD and 60 were healthy controls. Risk of bias assessment indicated that this study had a low risk of bias.

In contrast, the study that included participants with PDD found a weak negative to moderate positive correlation, $r = -0.06$ to 0.40 (Wolf et al., 2022). This was a prospective naturalistic intervention with 52 participants, of which 26 had PDD and 26 were borderline personality disorder controls. The study had a low risk of bias.

Similarly, the study with patients with various depressive and bipolar disorders found a moderate and strong association UOR = 2.8 and 7.07 (Wiktorsson et al., 2010). This was a case-control study with 511 participants, comparing 103 patients with affective disorders with a 408 comparison group.

Overall, these results highlighted that the association between loneliness and suicidality exists in patients with depressive disorders. In addition, depression may influence the association.

3.5. Comparison between general population samples and patients with depressive or bipolar disorders

Although there was a weak positive correlation between loneliness and suicidality in the study on BD patients (Pike, 2024), studies on enrolled general populations generally showed a moderate correlation. Therefore, BD may diminish the association between loneliness and suicidality relative to general population samples. In contrast, the study on PDD patients (Wolf et al., 2022) showed similar results to general population studies, in terms of correlation and odds ratio findings. Because of this similarity, PDD patients' experience of loneliness and suicidality may be more closely aligned with normative patterns, potentially suggesting a more stable emotional or cognitive state in relation to loneliness and suicidality. Although depression, loneliness, and suicidality are interconnected, the limited number of studies on depressive and bipolar disorders makes it difficult to confidently determine their role in the association between loneliness and suicidality.

3.6. Depressive symptoms as a mediator

Studies that specifically tested depressive symptoms as a mediator between loneliness and suicidality include McClelland et al. (2023), Lasgaard et al. (2011b), and Yang et al. (2023a).

McClelland et al. (2023) was a cross-sectional study rated fair in risk of bias assessment. The sample was drawn from healthy adults and older adults at the University of Glasgow. UCLA and the Social and Emotional Loneliness Scale for Adults (SELSA) were used to measure loneliness and the Suicide Probability Scale (SPS) for suicidality. Depressive symptoms were tested as a mediator of the association between suicidal ideation and SELSA-Family, SELSA-Romantic, and SELSA-Social loneliness. SELSA-Family measured perception of feeling disconnected or isolated from one's family, SELSA-Romantic assessed feelings of isolation or longing for a romantic partner, and SELSA-Social reflected feelings of detachment from one's broader social network. In SELSA-Family and suicidal ideation, depression partially mediated the association, consistent with UCLA and suicidal ideation. Depressive symptoms may

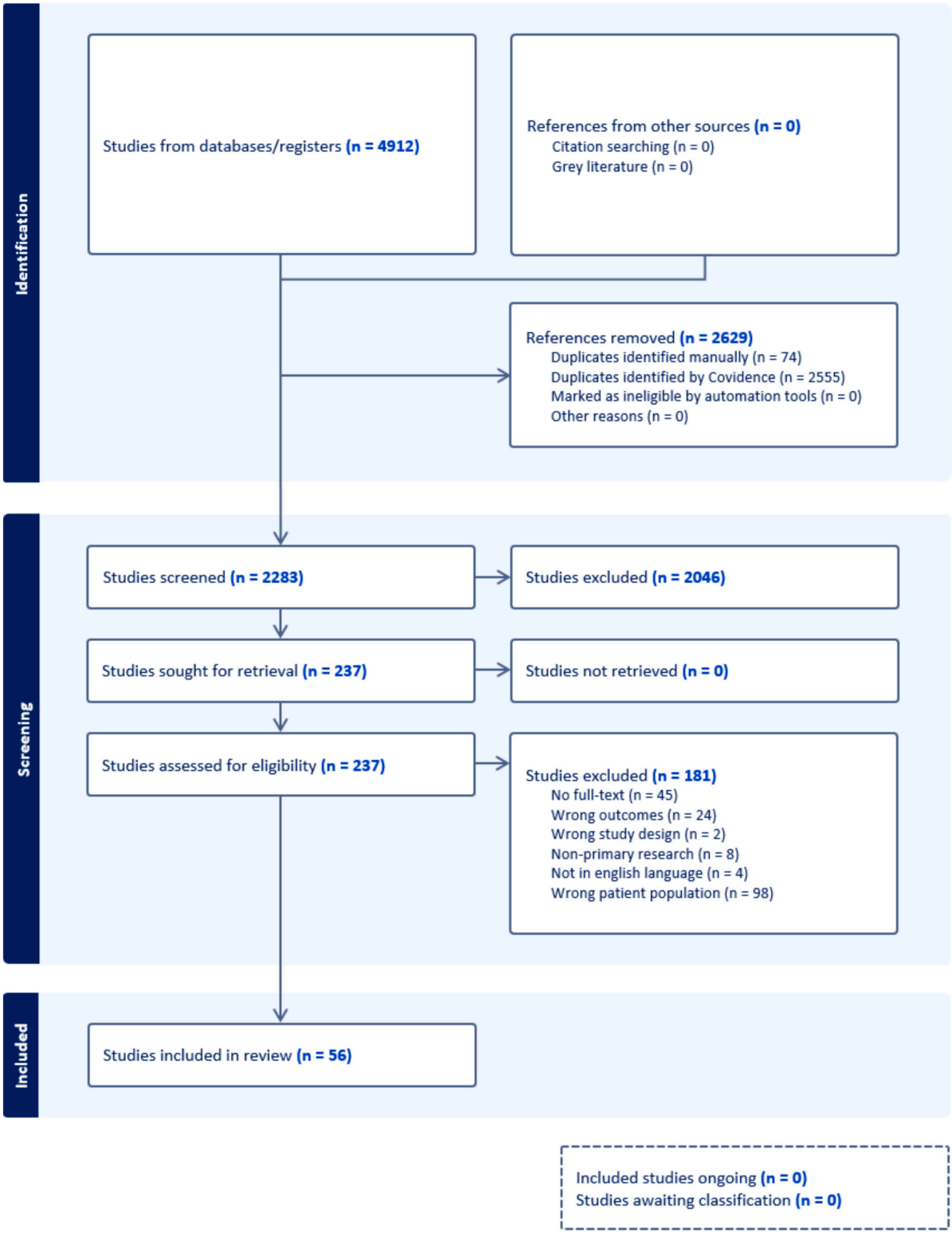

Fig. 1. PRISMA flow diagram of literature search (Covidence, 2024).

explain some of the association. However, in SELSA-Romantic and suicidal ideation, depression fully mediated the association as the association became non-significant after accounting for depressive symptoms. This suggests that depressive symptoms may entirely explain the link between romantic loneliness and suicidal thoughts. In SELSA-Social and suicidal ideation, depression did not mediate the association. This indicated that social loneliness may not be linked to suicidal ideation through depressive symptoms.

Lasgaard et al. (2011b) was a study that included a cross-sectional and observational cohort component. Only the cross-sectional component assessed the effect of depressive symptoms on the association between loneliness and suicidality. This study was rated fair in risk of bias assessment. The sample was drawn from adolescents and young adults from the High School Loneliness Study, a Danish national study. SELSA-Social was used to measure loneliness and the suicidal ideation subscale of SPS was used to measure suicidality. In hierarchical regression analysis, after adding loneliness and demographic variables in the association, depressive symptoms were added. The effect of loneliness on suicidal ideation dropped to non-significant and negative, suggesting that depression accounts for the association. Depressive symptoms appear to be a key mediator.

Yang et al. (2023a) was an observational cohort study rated good in quality assessments. The sample was drawn from high school students from Taizhou City, China. UCLA measured loneliness, while the Positive and Negative Suicide Ideation Inventory (PANSI) assessed suicidality. Three regression models were employed. Model 1 only included loneliness and suicidal ideation, while Model 2 adjusted for sociodemographic covariates and baseline suicidal ideation. In Model 3, after adjusting for baseline depressive symptoms, the association between loneliness and suicidal ideation became non-significant, suggesting that depressive symptoms accounts for the association.

Results from McClelland et al. (2023), Lasgaard et al. (2011b), and Yang et al. (2023a) provide evidence that depressive symptoms mediate the association between loneliness and suicidality, specifically suicidal ideation as this was the specific suicidality measure used. However, McClelland et al. (2023) found this to only be true for SELSA-Family, SELSA-Romantic, and UCLA loneliness, not SELSA-Social loneliness.

3.7. Other moderators in the association

Unique negative moderators that diminished the association between loneliness and suicidality include perceived stigma from bereavement (Pitman et al., 2020), jail-related stress (Bonner and Rich, 1990), and feelings of hopelessness (Joiner Jr and Rudd, 1996). Conversely, protective moderators include optimism (Chang et al., 2017a), school belonging (Zhou et al., 2023), and family support (Zhang et al., 2021). These findings suggest that a positive outlook, a sense of belonging at school, and support from family members may reduce the impact of loneliness on suicidality. However, Zhang et al. (2021) also found that support from friends and nursing staff did not significantly influence the association.

Ernst et al. (2021) noted that the association between loneliness and suicidal ideation was weaker in women than in men. Furthermore, a positive association between loneliness and protective factors against suicidal ideation was observed in transgender individuals, while no such association was detected among cisgender individuals (Yadegarfard et al., 2014). Notwithstanding, it is important to recognize that there are potential moderating factors other than age and depression, including but not limited to sex, economic status, education level, and access to social media (Park et al., 2020).

4. Discussion

4.1. Implications

Our findings indicate that loneliness exhibits a strong positive association with suicidality in general population samples, with age as a moderator. This association extended across those that were diagnosed with depressive disorders, including MDD and PDD. Studies that specifically tested depressive symptoms in the association between loneliness and suicidal ideation found that depressive symptoms typically fully mediated the association. However, McClelland et al. (2023) found that this mediation depended on the type of loneliness, as some types of loneliness were only partially or not all mediated in the association. These conclusions were in accordance with other analysis in this area insofar as associations between loneliness, disparate mental disorders, and suicidality outcomes have been reported (McClelland et al., 2020; Shaw et al., 2021; Schinka et al., 2012; Shoib et al., 2023). Results also align with the Interpersonal Theory of Suicide (ITS), where loneliness in the form of thwarted belongingness (feeling socially disconnected and lacking meaningful relationships) is identified as a risk factor to suicidal thoughts and behaviours (McClelland et al., 2020; Ribeiro and Joiner, 2009; Van Orden et al., 2010). In terms of BD, the magnitude of the association between loneliness and suicidality observed was less than what was estimated in general populations or depressive disorders. It remains unclear whether this finding reflects true associations or arises from reporting biases, differences in study design, sample characteristics, or measurement tools, or whether there is any aspect of living with BD that intrinsically or extrinsically alters risk in this context. Further research is required to clarify the role of bipolar disorders in the association between loneliness and suicidality.

Aforementioned moderators of the association between loneliness and suicidality, including age, highlight the importance of enhancing protective moderators and addressing negative moderators to reduce suicide risk (Park et al., 2020). There are also theoretical frameworks that explain the causal mechanisms between loneliness and suicidality and approaches to counteracting these pathways. Prolonged loneliness activates the hypothalamic pituitary-adrenal (HPA) axis, leading to increased stress hormone release, another risk factor for suicide (Park et al., 2020; Hawkley and Cacioppo, 2010; Miller et al., 2007; Wilkialis et al., 2021). This pathway is derived from persistent activation of stress responses, impairing emotional regulation and increasing mental health vulnerabilities (McEwen, 2004; Lupien et al., 2009). Furthermore, loneliness also disrupts normal reward and cognitive systems, changing dopaminergic pathways crucial for motivation and pleasure and thus increasing susceptibility to mood disorders like depressive and bipolar disorders (Cacioppo and Hawkley, 2009; Whitton et al., 2015). These cognitive alterations frequently appear as negative thought patterns and feelings of hopelessness that are interconnected with suicidal behaviour (Law and Tucker, 2018). Although loneliness increases stress and disturbs cognitive pathways, which increases risk for suicide, in contrast, social connections contributes as a buffer against stress, fostering resilience (Umberson and Karas Montez, 2010). In the absence of a social network, individuals can lose access to critical emotional support, further diminishing individuals' capacity to cope with life's challenges (Cohen, 2004). Other interventional approaches to these causal pathways include allopregnanolone (ALLO) to regulate HPA axis dysfunction and cognitive-behavioural approaches to reduce loneliness through addressing maladaptive thoughts (Park et al., 2020).

Our findings emphasize that prevention for loneliness is crucial. Preventions require scalable, accessible, implementable, cost-effective, appropriate strategies to reduce social isolation and loneliness across the general population (Goldman et al., 2024). The risk for loneliness and social isolation exists throughout all demographics but notably greater in younger and older populations (Blazer et al., 2020; Shah and Househ, 2023). In U.S. Surgeon General Vivek Murphy's review "Our Epidemic of Loneliness and Isolation: The U.S. Surgeon General's Advisory on the Healing Effects of Social Connection and Community" (Murphy, 2023), he presents various evidence-based recommendations to reduce loneliness in the general population. Governments can invest in social infrastructure at the local level that promote bringing people closer in proximity (Holt-Lunstad, 2023). In addition, financial

resources allocated to research on loneliness and social isolation can align with the severity and effects of these issues on society (Holt-Lunstad, 2023; National Academies of Sciences, Engineering, and Medicine, 2023). Health care systems and workers can integrate social connection into patient care through assessing patients' level of social connection, educating patients on benefits of social resilience, and using interventions that promote more interaction with their social network (Holt-Lunstad, 2023; National Academy of Medicine, 2024). Workplaces can create a culture that promotes connection between co-workers and establish policies that protect workers' ability to support their relationships outside work, including boundaries between work-hours and non-work-hours (Haas, 2023; Kossek et al., 2011). Furthermore, schools and educators can create a supportive school environment that fosters belonging and promotes collaboration through establishing rules and school-related activities (CDC, 2021; Roseth et al., 2008). In addition, education departments can develop a health curriculum that includes educating on the importances of social connections and maintaining a social network and the potential consequences involved when these protective factors are disrupted.

4.2. Strengths and limitations

The studies synthesized in this systematic review possessed several notable strengths. Various study designs were used, including cross-sectional, prospective observational cohort, and case-control. In addition, the inclusion of studies with various age groups, clinical populations, and diverse cultural and geographical backgrounds added to the robustness of the findings. The diversity of these study characteristics and designs with large sample sizes allowed for a comprehensive understanding of the association between loneliness and suicidality, strengthening the external validity of findings. Furthermore, a variety of analyses were employed that yielded odds ratios, correlations, multiple correlations, chi-square, beta-coefficients, and variance. These measurements also added robustness to the findings, demonstrating that loneliness and suicidality were associated across different statistical methods and assessments.

A significant limitation was the overwhelming inclusion of studies that used cross-sectional designs compared to prospective observational cohort designs and case-control, which limited the ability to make causal inferences. Although cross-sectional studies provided valuable insights, such as elucidating associations between loneliness and suicidality at a given point in time, they did not allow researchers to determine whether loneliness directly causes or leads to suicidality or vice versa. Moreover, the number of studies that examined the older adult population and depressive or bipolar disorder populations without comorbidities was low. Mixed age group studies also did not allow direct comparisons due to the production of weakened findings. Therefore, we mostly relied on studies that clearly distinguished between age groups (healthy adults, older adults, or adolescents) to provide more focused and relevant comparisons. In addition, we only separated groups based on age and depressive or bipolar disorders, considering only age as a moderator and depressive symptoms as a mediator, and the influence of depressive and bipolar disorders.

Some studies measured loneliness and suicidality using items created specifically for the study, not validated and reliable measurements. In addition, while most studies were rated fair in quality assessments, the few studies rated poor may have influenced the results of our study due to their higher likelihood for bias. Findings had limited generalizability due to the inclusion of significant and non-significant results and different approaches to suicidality measurement (suicidal ideation, attempt, plan, suicidal behaviour, suicide probability, etc.).

5. Conclusion

To our knowledge, this is the first systematic review to examine the association between loneliness and suicidality across different general population age groups and compare these findings with patients diagnosed with depressive and bipolar disorders, including MDD, PDD, and BD. McClelland et al. (2020) conducted a similar recent systematic review on loneliness as a predictor of suicidal ideation and behaviour, with insights on depression as a mediator of loneliness. Our results provide critical insights into the unique and overlapping risk factors for suicidality, suggesting the importance of tailored suicide prevention strategies that address loneliness in both clinical and non-clinical populations. Preventative measures are encouraged to consider age and depressive symptoms as moderators while enhancing protective factors and reducing negative factors. In addition, future research should prioritize longitudinal designs to further explore the association between loneliness and suicidality across time. Studies should employ standardized and validated measures of loneliness and suicidality to enhance comparability to other studies. Moreover, studies are encouraged to focus on underrepresented populations in the literature, such as older adults and patients with depressive or bipolar disorders, and consider other covariates in the association between loneliness and suicidality. Practitioners should routinely evaluate persons living with depressive or bipolar disorders for loneliness as part of a comphrensive assessment.

CRediT authorship contribution statement

Andy Lu: Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – review & editing, Writing – original draft. **Allyssa Chan:** Investigation, Data curation. **Trisha Menon:** Investigation, Data curation, Writing – review & editing. **Gia Han Le:** Methodology, Conceptualization. **Roger S. McIntyre:** Supervision, Writing – review & editing. **Poh Khuen Lim:** Writing – review & editing. **Hernan F. Guillen-Burgos:** Writing – review & editing. **Roger Ho:** Writing – review & editing. **Heidi Ka-Ying Lo:** Writing – review & editing. **Taeho Greg Rhee:** Writing – review & editing. **Sabrina Wong:** Writing – review & editing.

Declaration of competing interest

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2025.03.150.

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